MEASURING SCHOOL ECONOMIC DISADVANTAGE

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Abstract

Many educational policies hinge on the valid measurement of student economic disadvantage at the school level. Measures based on free and reduced-price lunch enrollment are used widely. However, recent research raises questions about their reliability, particularly following the introduction of universal free lunch in certain schools and districts. Using unique data linking the universe of students in Oregon public schools to IRS tax records and other data housed at the U.S. Census Bureau, we provide the first examination of how well different measures capture school economic disadvantage. We find that, in Oregon, direct certification provides the best widely-available measure, both over time and across the distribution of school economic disadvantage. By contrast, neighborhood-based measures consistently perform relatively poorly.

Keyword*:* School Lunch; School Economic Disadvantage; Measurement of Poverty; Validity; Economically Disadvantaged Students

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Introduction

We are living through an era of profound economic inequality and deep gaps in academic achievement between low-income and non-low-income students (Reardon 2011; Hashim et al., 2020; Bailey and Dynarski 2011; Duncan, Kalil, and Ziol-Guest 2017; Ziol-Guest and Lee 2016). [1] Long-standing efforts seek to improve the educational resources and opportunities available to economically disadvantaged students. Many such compensatory strategies, including Title I of the federal *Elementary and Secondary Education Act* as well as a variety of state and district-level programs, earmark supplementary funds and educational programs to schools and districts that educate relatively large proportions of economically disadvantaged students (Greenberg, Blagg, & Rainer, 2019; U.S. Department of Education, 2019). Oregon's state-level school finance system, for example, provides additional funding to students deemed economically disadvantaged by applying a multiplier of 1.25 to the base per-pupil amount for these students (State of Oregon, 2020). Similarly, California applies a multiplier of 1.20 to the base per-pupil amount for economically disadvantaged students and provides an additional grant for districts where at least 55% of students are economically disadvantaged (California Department of Education, 2021).

To operate efficiently and equitably, compensatory school funding programs require valid measures of the concentration of economic disadvantage at the district and school levels (Marar, 2020). However, substantial questions exist about the degree to which the most widely-used existing measure – the proportion of students enrolled in means-tested free or reduced-price lunch (FRPL) via the National School Lunch Program (NSLP) – can accurately capture

economic disadvantage in schools (Domina et al. 2018; Michelmore & Dynarski 2017; Harwell & LeBeau 2010).

In this paper, we use confidential data housed at the U.S. Census Bureau to analyze the validity of various measures of school economic disadvantage that are currently in use or proposed for use. We link federal tax return and program participation microdata to administrative public school enrollment data for the state of Oregon. We then construct a school economic disadvantage measure that, thanks to the breadth and accuracy of federal tax data, in particular, offers a benchmark against which we compare an assortment of candidate measures of school economic disadvantage. Specifically, the benchmark captures the proportion of students enrolled in each Oregon public school who were either enrolled in SNAP or whose family reported a household income of less than 185 percent of the federal poverty level on their tax return.

We use this benchmark to assess the following policy- and research-relevant candidate measures of school economic disadvantage: two different sources of FRPL enrollment rates, so-called direct certification rates (which measure the proportion of students who are enrolled in SNAP and other means-tested programs targeted at economically disadvantaged families, such as Medicaid or Temporary Assistance for Needy Families, depending on the specific state's data linking processes), and four experimental measures (one from the Urban Institute, one from the National Center for Education Statistics and one derived from the National Center for Education Statistics data, and one using aggregated data describing the rate of poverty in the neighborhoods in which students live). We posit that any measure used to distribute school finances aimed at compensating for school economic disadvantage ought to mirror variation in economic disadvantage rates over time and across schools relative to the benchmark in an unbiased

fashion. As such, our analyses compare temporal trends for our benchmark measure and available candidate measures; examine the cross-school correlations between these measures; and focus on the extent to which the benchmark and candidate measures correspond across the distribution of school economic disadvantage.

Our analyses reveal that, in Oregon, many widely available measures of school economic disadvantage correlate reasonably highly with our benchmark school economic disadvantage measure. However, each of these measures is subject to important shortcomings. We find that the validity of widely used FRPL enrollment rate data depends on how these data account for growing number of schools that participate in universal free meals programs, the most notable of which is the Community Eligibility Provision. By contrast, direct certification rates correlate highly with benchmark economic disadvantage rates, as they reflect participation in means-tested programs distinct from NSLP. Not all states make direct certification rates available for public use and fewer use them for accountability or funding policy purposes (see Ohio Department of Education, 2021). But our analyses suggest that, among measures with widespread availability, direct certification provides the most valid measure of economic disadvantage at the school level.

Other existing alternatives, including measures that use population-based estimates of household poverty rates or income levels of households living near students or their schools, are less promising. Such measures also preclude understanding which specific students are in poverty, which, though not a focus of our analyses, is an important feature for research efforts that investigate or attempt to control for within-school variation in economic status, including, as examples, the literature on peer effects (see Sacerdote, 2011), within-school income segregation (see Dalane and Marcotte, 2022), and teacher effects literatures (see Jackson, Rockoff, Staiger, 2014).

Background

Defining the construct: "Students who are economically disadvantaged"

Our focus in this paper is on the valid measurement of student economic disadvantage rates at the school level. While other measures of student disadvantage and academic need are clearly important, we focus on economic disadvantage (and the closely related constructs of poverty and low-income) because of the central role that this construct has long played in U.S. educational policy. Over the last several decades, policymakers at the federal, state, and local levels have authorized a wide array of policies that direct supplementary funds to public schools and districts based on the concentration of economically-disadvantaged students. These policies conceptualize economic disadvantage as a unidimensional and dichotomous construct – a student is either economically disadvantaged or not, based on their household income. In line with this policy tradition, we focus largely on school economic disadvantage measures constructed from dichotomous indicators based on household income.

The largest and most influential policy aimed at providing additional support for economically-disadvantaged students is Title I, Part A of the 1965 *Elementary and Secondary Education Act* (ESEA). Title I allocates supplemental federal educational funds to "local educational agencies serving areas with concentrations of children from low-income families to expand and improve their educational programs." The policy's language notes the "special educational needs of children of low-income families and the impact that concentrations of low-income families have on the ability of local educational agencies to support adequate educational programs" (Public Law 89-10-Apr. 11 1965, p. 27).

Currently 35 states also provide additional funding for pre-primary, elementary, secondary, and higher education targeting economically disadvantaged students (Edbuild, n.d.a).

Furthermore, many of the nation's largest districts have implemented weighted student funding formulae that inform the distribution of district resources across schools. Although existing district-level systems consider a range of student attributes, including the concentration of English Language Learners and special education students, most include student poverty as a key factor for targeting compensatory resources (Roza, Hagan & Anderson 2021; Edbuild, n.d.b). These federal-, state-, and district-level policies vary in many regards. Different policies operationalize the notion of economic disadvantage or poverty in different ways, with some emphasizing the federal poverty line and others emphasizing different income-based thresholds. Policies also differ in the degree to which they supplement allocations based on student poverty or other forms of economic disadvantage and the range of other characteristics that inform resource allocations (EdBuild, n.d.b; Roza, Hagan & Anderson 2021). Nevertheless, common to each of these policies is the notion that school finance mechanisms can offset disparities associated with socioeconomic inequality by directing funds toward the schools and districts that educate students from economically disadvantaged families.

Measures of school-level economic disadvantage

In this section, we provide a general overview of strategies currently being used for the measurement of student economic disadvantage at the school level in U.S. policy contexts as well as several promising alternative strategies. Detailed information on the specific measures evaluated is provided in a later section.[2]

FRPL enrollment

To measure economic disadvantage at the school level, policymakers typically rely on data produced in the administration of means-tested programs designed to benefit children living in economically disadvantaged households. Historically, the most widely-used measure is based

on the proportion of students enrolled in FRPL. Currently approximately 30 U.S. states use FRPL data to measure school economic disadvantage for funding purposes (EdBuild, n.d.b).

Scholars and practitioners have long known that FRPL rates provide error-prone measures of school economic disadvantage (Domina et al. 2018; Harwell and LeBeau, 2010). Historically, students enroll in FRPL when their households self-report income at or below 185 percent of the federal poverty line. This self-reporting process produces two-sided error. Some students from families with incomes above 185 percent of the federal poverty line enroll in FRPL; other students from economically disadvantaged families do not enroll (Domina et al. 2018). The measurement error in the FRPL rate likely varies across schools and communities with variation in outreach, as well as concerns about being stigmatized as poor at school and governmental distrust (USDA, 2022, Richman, 2019, Domina et al. 2018; ERS, 2006; Harwell and LeBeau, 2010).

Recent efforts to improve student access to school meals have further exacerbated concerns regarding the ability of FRPL data to measure economic disadvantage (Cookson, 2020; Koedel & Parsons 2021). Specifically, Provisions 1, 2, and 3 of the National School Lunch Program (implemented in the 1980s as alternative provisions to the normal requirements for annual determinations of eligibility for FRPL) and the Community Eligibility Program (CEP, available nationally in 2015) permit schools to provide free lunch widely, and directly or indirectly reduce data collection obligations. During the 2019-20 school year, approximately 69 percent of eligible schools participated in CEP (FRAC, 2020). Some schools continue to report traditional FRPL enrollment data after enrolling in CEP, while others stop reporting or report that all students are enrolled (Greenberg et al. 2019). The pandemic potentially brought additional changes to FRPL enrollment data as schools struggled to collect forms from eligible families and

federal pandemic relief efforts allowed schools across the country to offer free meals to all students beginning in the spring 2020 (Burnette II, 2020). While the pandemic relief effort ended in the fall of 2022, several states have passed legislation ensuring that all students maintain access to free school meals (Belsha, 2023). In the wake of these changes, it is not clear when and to what extent schools and districts will resume FRPL data collection.

Direct Certification

In the 1990s and early 2000s, the USDA took steps to simplify the process for families and schools to enroll children in FRPL by mandating that states create data systems linking education records with records from SNAP and other means-tested programs using student personally identifiable information. These data linkages allow students already enrolled in means-tested programs to be directly certified for FRPL, reducing the burden on individual household applicants to complete eligibility forms and on program administrators to process applications. More recently, some five states piloted a program to incorporate Medicaid data for direct certification (Hulsey et al., 2015) in the 2012-2013 school year. This program continues to expand, with some 14 additional states – including Oregon - planning to incorporate Medicaid to the direct certification process in the 2023-2024 school year for both free and reduced-price meal eligibility (USDA, 2023).

Many states and school districts now use the proportion of students who enroll in FRPL via direct certification either as a supplement to or a replacement for traditional FRPL rates for resource allocation. To date, limited evidence is available regarding the properties of direct certification as a measure of school economic disadvantage. These data are not strictly comparable to FRPL rates, since the criteria for receiving benefits via SNAP and TANF are more restrictive than the criteria for enrolling in FRPL directly.[3] Thus, direct certification rates have

both potential strengths and weaknesses relative to NSLP data as a measure of school economic disadvantage. Since families must provide income documentation to enroll in SNAP and TANF, direct certification rate data may be less subject to some forms of bias than FRPL enrollment data. At the same time, the policies and administrative practices that govern the process of enrolling in SNAP, TANF, and Medicaid vary across states, within states, and over time (Ganong and Liebman, 2018; Dickert-Conlin et al., 2021). This variation likely leads to both geographic and temporal variation in the characteristics of families who are captured in direct certification. In particular, direct certification rates may substantially understate economic disadvantage rates in contexts where program enrollment processes place heavy burdens on potential beneficiaries relative to contexts where administrative burdens associated with program enrollment are less pronounced (Herd & Moynihan 2018; Heinrich et al. 2022). Direct certification data may also understate economic disadvantage in schools serving immigrant, rural, and other communities in which income-qualified families are relatively less likely to enroll in means-tested programs.

Alternative candidate measures

In order to address the shortcomings associated with measures of economic disadvantage based on program participation data, researchers and policymakers have begun to experiment with alternative approaches. Some states and districts construct school-level measures of student neighborhood economic disadvantage using ACS data (see Morath, 2019 for an example in Texas and see Maryland State Department of Education, 2022). While these data do not directly measure the household income of students enrolled in schools, they use student home address data to identify the Census block groups or tracts in which students live and use publicly-available data on neighborhood economic conditions to proxy for students' own economic circumstances. If students' economic circumstances correlate highly with those of their

neighbors, aggregating these measures to the school level may accurately approximate rates of school economic disadvantage.

The Urban Institute recently developed a measure of school economic disadvantage that builds on two sources of publicly available data: the school-level direct certification or FRPL data available via the National Center of Education Statistics' Common Core of Data (CCD) and the Census-constructed district-level Small Area Income and Poverty Estimates Program (SAIPE) (Gutierrez, Blagg, and Chingos, 2022). This measure, known as Model Estimates of Poverty in Schools (MEPS), uses a linear mixed effects model to predict school-level estimates of economic disadvantage based on average household economic disadvantage rates for families with children in the districts in which schools are located. MEPS makes two assumptions as it attempts to address error associated with FRPL or direct certification data: (1) that school-level biases in FRPL or direct certification data cluster at the district level (as opposed to varying across schools within the same district), and (2) that SAIPE data capture public school student poverty rates at the district level, despite being based on a sample that is not designed to specifically generalize to the subset of families enrolling children in public schools. SAIPE data captures the proportion of school-aged children below 100 percent of the federal poverty line within a school district, which is a lower threshold of poverty than other typical definitions. Because of this, MEPS identifies fewer students as economically disadvantaged compared with other measures.

The National Center for Education Statistics (NCES) reports School Neighborhood Poverty (SNP) statistics. This measure uses ACS data to estimate the average household income relative to the poverty line for the households with school-aged children who live in close physical proximity to each school's campus (Geverdt and Nixon, 2018). Fazlul et al. (2021)

transform the SNP measure designed to better match the dichotomous conception of poverty embedded in policy, capturing the share of students at or below 130 percent of the federal poverty line for each school. To the degree that the economic circumstances of households located near a school's campus are representative of the economic circumstances of all the households in which enrolled students live, these SNP-based measures can provide a proxy for school economic disadvantage (Fazlul et al. 2021). However, this may be a problematic assumption, particularly for schools that serve geographically dispersed student bodies.[4]

Current study

Our analyses investigate the validity of various measures of student economic disadvantage at the school level, with the goal of identifying measures that are both reliable and widely available for use in educational policy and practice. Our argument-based approach to validation (Kane, 2006, 2012) proceeds by establishing a benchmark measure and then assessing alternative measures against our benchmark.

Specifically, our analyses address the following research questions:

1. To what degree do available measures of economic disadvantage track temporal trends in economic disadvantage for students enrolled in Oregon public schools?

2. To what degree do available school-level measures capture cross-sectional variation in economic disadvantage in Oregon public schools, and how has this changed over time?

3. To what degree do available measures capture variation across the distribution of school economic disadvantage? In particular, do candidate measures over- or under-estimate school economic disadvantage rates in schools enrolling high or low concentrations of students from economically disadvantaged families?

Data and measures

We link student-level administrative records from the Oregon Department of Education (ODE) for the school years 2009-10 through 2016-17 with administrative records stored at the U.S. Census Bureau. Students are linked to their Oregon SNAP enrollment records for each school year, and the IRS form 1040 records (tax returns) in which they are claimed as dependents for the most recent and two prior calendar years (spanning from 2007 to 2016). These linked data allow us to construct two benchmark measures of student economic disadvantage, which we aggregate to the school-by-year level for this paper's analyses. We compare these two benchmarks against seven more broadly available candidate measures of economic disadvantage, which we describe in detail below and in Table 1.

We use protected identification keys (PIKs) assigned via the Census's Person Identification Validation System to link across data sources.[5] The overwhelming majority of students receive PIKs: in the 2016-17 school year, we link over 95% of K-12 students in the ODE data to PIKs. If a student does not receive a PIK, we cannot link them to IRS and SNAP data. We exclude students without PIKs from our calculations of school-level economic disadvantage rates. Broadly speaking, students might not be assigned a PIK due to missing or erroneous information in ODE administrative records for the student, or because the student's information is not present in the reference file used to assign PIKs, which means a PIK has never been created for them. By excluding students without PIKs, we make an assumption that students are missing income at random at the school-level: that is, we assume that students whose economic disadvantage we do not observe due to missing PIKs have, on average, the economic disadvantage rate of the school. The same missing income at random assumption is used for students who have PIKs but do not appear in IRS 1040 records. Clark and Bhaskar (2023)

document that this assumption is more plausible than assuming students with missing values are categorically economically disadvantaged.

Benchmark Measures

Benchmark (SNAP/IRS 3 year)

Our benchmark measure of school economic disadvantage defines students as economically disadvantaged in a given year if they meet one of two tests: SNAP enrollment or family income less than 185 percent of the federal poverty line. Students are considered economically disadvantaged via SNAP receipt if they were enrolled in SNAP at any time during the school year. For students who were not enrolled in SNAP, the income determination is made using IRS 1040 records from the most recent calendar year (e.g., 2016 IRS records for the 2016-17 school year). In preliminary work, Clark and Bhaskar (2023) find that Hispanic children and children living in non-married households are less likely to appear in IRS 1040 records, but incorporating SNAP data virtually eliminates this bias.

Specifically, we identify students as dependents in their families' tax returns and calculate their ratio of income to poverty by dividing their adjusted gross income by the poverty line for their family size (also derived from the 1040). For students who are not enrolled in SNAP or claimed on a 1040 in the most recent calendar year, we check 1040 records from the prior two calendar years, and use this income information. In summary, we deem the following students economically disadvantaged in the 2016-17 school year: a student enrolled in SNAP at any time during the school year regardless of their family's tax-reported income; a student not enrolled in SNAP whose family's 2016 income-to-poverty ratio is below 185 percent of the federal poverty line; and a student not enrolled in SNAP and not linked to a 2016 tax return, but

whose family met the income standard in their 2015 tax return (or, if that is also missing, their 2014 tax return).

Eight percent of Oregon students in the 2016-17 school year are missing this measure. As with all the microdata-derived measures used in this paper, we use listwise deletion for these missing cases, implicitly assuming they are missing-at-random at the school level. This assumption holds that students with missing income have, on average, similar rates of economic disadvantage to their school peers with observed income.

We interpret this benchmark as a measure of the proportion of students who are economically disadvantaged, with 185 percent federal poverty as our target threshold.[6]

Benchmark (IRS 1 year)

We prefer a benchmark that incorporates SNAP data, since it improves coverage and reduces bias. However, it is additionally useful to have a benchmark that is not mechanically correlated with our SNAP-derived direct certification measure. Thus, we construct an alternative version of our benchmark measure that uses only IRS 1040 data on family income and size from the most recent calendar year to measure the proportion of students with family income below 185 percent of the federal poverty line. This measure systematically excludes students whose families did not recently file tax returns, but it offers a measure that is independent of program participation, and one that states may be able to more easily replicate with state level tax data. 15.4% of students are missing on the Benchmark (IRS 1 year) measure. The Benchmark (IRS 1 year) measure correlates with our preferred Benchmark (SNAP/IRS 3 year) at 0.99. This correlation suggests that the missing-at-random assumption is effective for deriving school-level economic disadvantage rates using tax data alone despite its imperfect coverage.

FRPL Enrollment

Two distinct measures of school-level economic disadvantage rates based on FRPL enrollment are currently available for Oregon public schools. Our analyses use both, since they highlight potentially important differences in how FRPL rates address measurement questions associated with CEP and other programs that offer free school meals to all students at participating schools.

CCD FRPL

The NCES Common Core of Data (CCD) collects FRPL rates for Oregon public schools, which we refer to as "CCD FRPL". NCES state coordinators responsible for submitting school economic disadvantage information to the CCD receive guidance to "*estimate the count of students by multiplying current year membership by the percentage of eligible students in the most recent year for which the school collected that information" (U.S. Department of Education, 2021; p.10)*. Thus, since the introduction of CEP, some schools enrolled in CEP have continued to report their pre-CEP FRPL rate to the CCD in lieu of an updated rate.

ODE Economic Disadvantage

The ODE data flags students as "economically disadvantaged" based on their FRPL enrollment and other factors, including homeless children, as determined by the McKinney-Vento Homelessness Assistance Act (enacted in 2001). In 2015, certain schools participating in CEP or other universal meals programs show at or near 100 percent of students as "economically disadvantaged". Thus, our ODE Economic Disadvantage variable is the proportion of students in each school designated economically disadvantaged in the ODE microdata.

Direct Certification

SNAP

Although many states have started publishing school-level direct certification rates, Oregon did not during the time period of study. Thus, we construct an estimate of the direct certification rates for Oregon schools using Oregon SNAP data housed at the U.S. Census Bureau as part of the *Census-FNS-ERS Joint Project*, a cross-agency collaboration between the State SNAP agencies, the U.S. Census Bureau, and USDA's Food and Nutrition Service (FNS) and Economic Research Service (ERS).[7] This measure is simply the proportion of students at each school that are enrolled in SNAP at any point during the school year.

While we consider this measure of SNAP enrollment a proxy for direct certification, we note that different states use different means-tested programs – and typically more than one – to directly certify students. They also have different methods for matching students to direct certification records. [8] In Oregon, Medicaid will be added to direct certification beginning in the 2023-2024 school year. We are not able to assess the quality of Oregon's matching approach nor the effect of the introduction of Medicaid on the quality of the approach.

Alternative Candidate Measures

ACS Block Group

We construct an ACS-based measure by linking students in each school to their residential Census block group, then assigning them the economic disadvantage rate (proportion of households with income less than 185 percent of the federal poverty line according to the ACS 5-year estimate) of their block group.[9] We average these rates across students to arrive at a school-level measure of economic disadvantage.

Urban Institute MEPS

We utilize the publicly available measures from the Urban Institute's website, including the Model Estimates of Poverty in Schools (MEPS) and the modified MEPS. We follow the

Urban Institute's suggestion and report findings from the original MEPS in our main analyses and provide information about the modified MEPS in footnotes.

NCES-SNP

As described above, SNP measures the household income-to-poverty ratio for ACS respondent households living near schools. We include this measure for the year 2016-2017.

Transformed NCES-SNP.

Fazlul et al (2021) propose a transformation of the SNP designed to better match the dichotomous conception of economic disadvantage embedded in policy and other widely used measures. We reproduce this transformation and report results of this transformed variable as well as the untransformed version.[10]

[TABLE 1 AROUND HERE]

Sample of Schools

Table 2 provides descriptive statistics for the analytic sample. To construct a panel of schools comparable across all years of our study, we restrict the sample to schools (a) that appear in ODE administrative data in all 8 years of our study, and (b) for which all our candidate measures are available. All the analyses in this paper are restricted to the approximately 1,060 schools in our analytic sample. We lose approximately 300 schools with small enrollments driven in large part by instances for which there is no school-level data available in NCES data sources like the CCD and SNP. Our analyses consider only measures that are widely available throughout the state, but we note that some measures are not available for all schools. There are a small number of low-enrollment Oregon public schools in which no enrolled students are linked to data with which we construct the benchmark measure (they may be missing a PIK or, if not, do not appear in form 1040 or SNAP records). While the analytic sample offers good coverage of

the universe of schools in the ODE data (see Appendix Table 1), we note that the capacity of measures to cover all schools could be an important consideration from a policy perspective since missing schools could restrict education administrators' efforts to allocate compensatory funds using certain measures.

[TABLE 2 AROUND HERE]

Methods

To answer our first research question, we plot state-wide trends in school economic disadvantage for Oregon public schools on all available measures between 2010 and 2017.

To answer our second research question, we correlate all the candidate school economic disadvantage measures with our Benchmark (SNAP/IRS 3 year). Correlation coefficients indicate the extent to which the various measures move together, and how well each preserves the relative rankings of schools according to our benchmark. Our preliminary analyses focus on correlations in measurements drawn from the 2016-17 academic year. For the subset of measures that are available for three or more years, we further consider secular trends in measurement properties by estimating correlations separately for each school year, in light of concerns about the way changes in policy and practice induce change in proxy measure performance. Correlations are unaffected by whether the school economic disadvantage measure targets students at 100, 130, or 185 percent of the federal poverty line.

While informative, correlation analyses may overlook systematic divergences between measures in very low- or high-income schools. Our third research question draws attention to the performance of candidate measures across the distribution of school economic disadvantage. We focus on our most recent year of data (2016-2017) to address this question. We standardize each of our measures, group schools into 20 quantiles based on their rank according to the Benchmark

(SNAP/IRS 3 year) measure, calculate the mean and standard deviation of the candidate measures for schools in each quantile, and plot these binned statistics. These quantiles are enrollment-weighted, so each bin is comprised of schools enrolling approximately 5% of students. Each point on this plot represents the mean for the candidate measure in each quantile bin, providing a visual summary of how consistently the benchmark and candidate measures agree about the relative economic disadvantage of schools. Each of the whiskers represents the standard deviation of the candidate measures in each quantile bin, providing a visual summary of the extent of variation in the candidate economic disadvantage rates in each bin.

Results

Temporal trends in measures of student economic disadvantage

Figure 1 provides an overview of each of the candidate measures of economic disadvantage in Oregon public schools between the 2009-10 and 2016-17 school years. Since these measures are enrollment-weighted, they can be interpreted as the proportion of Oregon public school students designated as economically disadvantaged. Since the six measures represented on this graph use different income thresholds to define student economic disadvantage, it is not surprising that these lines depict different levels of economic disadvantage. However, we generally expect the measures to follow similar trends; as the share of students receiving SNAP falls, so too should the share enrolled in FRPL and the share designated as economically disadvantaged by MEPS. We find that all the candidate measures generally follow similar trends in economic disadvantage as the Benchmark (SNAP/IRS 3 year), with the exception of the ODE Economic Disadvantage measure.

[FIGURE 1 AROUND HERE]

The solid line near the top of the graph represents the Benchmark (SNAP/IRS 3 year) measure, which is the statewide trend in the proportion of Oregon public school students whose family reported household income less than 185 percent of poverty in IRS filings and/or enrolled in SNAP. This rate increases modestly at the beginning of the study period as the state slowly emerged from the 2008 recession, peaking at approximately 52 percent before declining to approximately 47 percent in 2017.

The dashed line with the circle marker in this graph represents the trend in CCD FRPL, as reported by the state of Oregon to the Common Core of Data; the dashed line with a triangle marker represents trends in the ODE Economic Disadvantage measure. The contrast between these two measures – both of which are ostensibly built around FRPL enrollments – is instructive. While the CCD FRPL rate follows year-to-year changes in the Benchmark (SNAP/IRS 3 year) measure, the ODE Economic Disadvantage rate – which counts schools that participate in CEP as 100 percent economically disadvantaged – rises as schools across the state implement CEP. The apparent reliability of the CCD FRPL measure may be deceptive: CEP schools have the option of reporting their pre-CEP FRPL rates to the CCD. Since school rates tend to be similar over time this creates a reasonable approximation, but it does not conceptually meet the needs of administrators and researchers because rates of economic disadvantage do not change over time in the schools that are using data from prior years.

As expected, the remaining measures – SNAP, ACS Block Group, and Urban Institute MEPS – record very different levels of economic disadvantage than the Benchmark (SNAP/IRS 3 year). SNAP enrollment rates, which reflect students whose families demonstrate household income below 130 percent of poverty, vary between 32 and 37 percent across the time period; ACS Block Group rates range similarly from 30 to 35 percent; and the Urban Institute MEPS

rate, which reflects students whose families have household income below 100 percent of poverty, varies between 20 percent and 15 percent in the years for which it is available. Trends in economic disadvantage rates using these measures follow the general economic disadvantage rates of the Benchmark (SNAP/IRS 3 year) over time.

Correlations between the benchmark measure and candidate measures

Table 3 summarizes the school-level correlations between the Benchmark (SNAP/IRS 3 year) and each of the two traditional FRPL measures, the direct certification measure, and the four alternative candidate measures described above. All measures in Table 3 pertain to the 2016-17 school year, the most recent year for which we have data. Our analyses indicate that the CCD FRPL measure correlates well with the benchmark in 2017, at 0.934, but since many schools are using data from previous years, we expect that the quality of this measure will deteriorate over time. The ODE Economic Disadvantage measure, which is largely based on FRPL enrollment and is affected by school participation in universal meals programs, has a lower correlation with the benchmark of 0.863. The difference in the correlation between the two traditional FRPL measures and the Benchmark (SNAP/IRS 3 year) indicates that the expansion of access to free meals differentially affects the quality of FRPL enrollment measures, depending on the decisions that data providers make as they construct and report data.

The analyses in Table 3 show that the direct certification measure, constructed using SNAP enrollment, correlates extremely highly with the Benchmark (SNAP/IRS 3 year) at 0.957.[11] It also is the measure with the highest correlation with the Benchmark (IRS 1 year) measure, confirming that its strong performance against the Benchmark (SNAP/IRS 3 year) measure is not mechanical. This suggests SNAP-based direct certification measures have strong potential as measures of school economic disadvantage.

Concerns about the measurement quality of school economic disadvantage measures based on FRPL enrollment have also led some to consider the use of alternative measures. Our ACS Block Group measure, which describes the degree of economic disadvantage in the neighborhoods in which schools' students reside, correlates with the Benchmark (SNAP/IRS 3 year) at 0.796. The Urban Institute MEPS, which is based on CCD, correlates with the Benchmark (SNAP/IRS 3 year) as highly as the CCD FRPL measure at 0.935.[12] Meanwhile, the NCES-SNP measure, which captures the economic disadvantage of families living in schools' immediate vicinity, is -0.694. The negative correlation is expected because the measure is an average of families' income-to-poverty ratios and thus should be inversely related to proportion low income. A Transformed NCES-SNP measure (cf, Fazlul et al. 2021) has a correlation of 0.661. Taken together, the results in Table 3 indicate that alternative candidate measures that use data from the students' ACS Block Group correlate less well with the Benchmark (SNAP/IRS 3 year) than do more readily available SNAP or FRPL enrollment-based measures.[13]

[TABLE 3 AROUND HERE]

Figure 2 graphs correlations between the Benchmark (SNAP/IRS 3 year), traditional FRPL-based measures, direct certification measure, and alternative candidate measures over time. We see that the quality of traditional FRPL-based measures varies over time. The correlation between the Benchmark (SNAP/IRS 3 year) and the CCD FRPL measure improves across the study period, while the correlation between the benchmark and the ODE Economic Disadvantage measure rises and then falls over time. While we do not have an explanation for the improvement that we observe in the ODE Economic Disadvantage measure's correlation with the Benchmark (SNAP/IRS 3 year) between 2010 and 2014, we note that the post-2014 decline

corresponds with the CEP's implementation in schools across the state. Some 39% of eligible schools adopted CEP beginning in the 2014-2015 academic year (Neuberger et al., 2015).

Encouragingly, this figure indicates that our proxy for direct certification – the measure of the proportion of students enrolled in SNAP (the dashed line with the square marker) – consistently correlates extremely highly with the Benchmark (SNAP/IRS 3 year) in each of the available years. If anything, this measure appears to improve in quality over time, with a correlation of .944 in 2010 and .957 in 2017.

We note that although the ACS Block Group measure does not correlate particularly highly with the benchmark at any point in the study period, its correlation is relatively stable over time. While we only have access to data on the Urban Institute MEPS measure for the last five years in the study period, we note that this measure appears to be somewhat more stable than both traditional FRPL enrollment-based measures.

[FIGURE 2 AROUND HERE]

Michelmore and Dynarski (2017) suggest flagging students as economically disadvantaged if they were economically disadvantaged for the prior 3 years. Since this requires multiple observations of the same student, it may not be a practical measure of year-to-year school-level economic disadvantage. It is nonetheless instructive to observe how highly correlated it is with our Benchmark (SNAP/IRS 3 year). We construct this measure of persistent economic disadvantage measure by flagging students as economically disadvantaged if they are indicated as such by the ODE Economic Disadvantage flag for the current school and the two years prior. The school-level rate of persistent economic disadvantage correlates at .835 with the Benchmark (SNAP/IRS 3 year) in 2016-17. Generally speaking, the correlation of the persistent measure with the benchmark is lower than that of the single-year ODE Economic Disadvantage

rate, with the exception of 2014-15 and 2015-16, where it briefly outperforms the single-year rate. We suspect this reflects the fact that the persistent measure holds onto the higher-quality pre-CEP data through 2015-16, improving its reliability relative to the single-year ODE Economic Disadvantage rate. However, once it is limited to data from after the introduction of CEP in 2016-17, it returns to slightly underperforming the single-year rate as a measure of school economic disadvantage.

Relationship between benchmark measure and candidate measures across the distribution

Figure 3 presents a series of binned scatterplots depicting the relationship between the benchmark (x-axis) and candidate measures (y-axis) across the distribution of school economic disadvantage. These graphs represent data from the 2016-17 academic year. We use z-scores to standardize all measures thus effectively setting all measures on the same scale, allowing us to focus on variation in measure rankings, rather than variation in levels of economic disadvantage. For purposes of legibility, we separate these scatterplots into panels: Panel A depicts the traditional FRPL-based measures, Panel B depicts the direct certification measure (SNAP), and Panel C depicts the ACS Block Group and Urban Institute MEPS measures. Each point on these graphs represents a bin of 5 percent of schools (approximately 55 schools).[14] In the graphs, we plot bin averages of the standardized candidate measure of school economic disadvantage (on the y axis) for each bin of the standardized Benchmark (SNAP/IRS 3 year) measure (on the x axis). Higher values on both axes indicate higher levels of economic disadvantage and lower values indicate lower levels of economic disadvantage. We also mark one standard deviation above and below the mean of the binned Benchmark (SNAP/IRS 3 year) measure, depicting how much variation each candidate measure has at different levels of the Benchmark (SNAP/IRS 3 year) measure.

[FIGURE 3 AROUND HERE]

The graphs in Panel A investigate the correspondence between the traditional FRPL-based measures and the Benchmark (SNAP/IRS 3 year), with the graph on the left focusing on the CCD FRPL measure and the graph on the right focusing on the ODE Economic Disadvantage measure. These traditional FRPL-based measures correspond closely with the benchmark across the distribution, although CCD FRPL measures school economic disadvantage with greater precision across the distribution than the ODE Economic Disadvantage measure. At the high end of the economic disadvantage distribution, these measures appear to understate economic disadvantage relative to the Benchmark (SNAP/IRS 3 year) measure, and this is particularly true for the ODE Economic Disadvantage measure. This is due in large part to the fact that the ODE Economic Disadvantage measure reports 100 percent of students enrolled in many CEP schools as FRPL - in 2017, 23 percent of schools in Oregon report a proportion of economically disadvantaged students greater than .99, while in the CCD FRPL measure only one school reports a proportion economically disadvantaged greater than .99. Because the ODE Economic Disadvantage measure reports many CEP schools at the scale's ceiling, this measure does not distinguish between those schools, and this dramatically alters the relative rank of many schools with relatively high levels of economic disadvantage.

While the graph in Panel B indicates that the direct certification proxy (SNAP) corresponds closely with the Benchmark (SNAP/IRS 3 year) across the bulk of the distribution of school economic disadvantage, we note that the SNAP measure captures school economic disadvantage rates less precisely at the very top of the distribution compared with other points of the distribution. This suggests heterogeneity in SNAP enrollment in more economically disadvantaged schools. Our data do not allow us to definitively account for why this might be,

but SNAP under-enrollment in economically disadvantaged schools is likely due to a relatively large concentration of children whose families qualify for SNAP benefits but choose not to enroll, a phenomenon that is particularly well-documented in immigrant communities (FRAC, 2020).[15] Additionally, the Food Distribution Program on Indian Reservations (FDPIR) may take the place of SNAP enrollment in schools near and in reservations. At the bottom of the distribution, we observe higher levels of the SNAP measure than expected based on the Benchmark (SNAP/IRS 3 year). This finding suggests that the relatively few students from economically disadvantaged families in economically advantaged schools have high levels of access to SNAP benefits.[16]

Looking at the alternative candidate measures in Panel C, we see that the ACS Block Group measure overstates economic disadvantage at the low end of the distribution and understates economic disadvantage at the high end of the distribution. Further, the ACS Block Group measure is relatively imprecise across the distribution of school economic disadvantage compared to other available measures. Finally, the Urban Institute MEPS corresponds to the benchmark measure similarly to the CCD FRPL measure, which makes sense given that Urban Institute MEPS relies on CCD FRPL enrollment data. Like the CCD measure, the Urban Institute MEPS understates economic disadvantage in highly economic disadvantaged schools. Unlike the CCD FRPL measure, the Urban Institute MEPS also slightly understates economic disadvantage in schools with relatively few economically disadvantaged students.

Conclusion

As the distribution of income becomes increasingly unequal, a broad array of federal, state, and local policies aim to target resources and opportunities toward students from

economically disadvantaged families and the schools that educate them. Valid measures of school economic disadvantage are an essential tool for the realization of this policy agenda.

Our analyses compare several widely used measures of school economic disadvantage as well as recently proposed alternatives against a benchmark measure constructed by linking student-level administrative data from the state of Oregon with IRS and SNAP data housed at the U.S. Census Bureau. Our findings build on a growing body of research highlighting the shortcomings of measures that rely on FRPL enrollment information. Using data from Oregon public schools, we demonstrate that the proportion of students who qualify as economically disadvantaged according to the state's official measure begins to diverge from our Benchmark (SNAP/IRS 3 year) economic disadvantage measure with the launch of when the NSLP's Community Eligible Provision in 2014-15. Trends in FRPL enrollment data reported in the CCD continue to correspond to the Benchmark (SNAP/IRS 3 year), though that may be due to CEP schools reporting pre-CEP FRLP rates to the CCD. These CCD-reported FRPL data correlate highly with our benchmark measure throughout the 2010 to 2017 period for which data are available, although we find some evidence to suggest that they understate the concentration of economic disadvantage in the state's most economically disadvantaged schools. While this may create the appearance of a reliable measure, we suspect that this measure will degrade over time. Notably, beginning in 2019-2010, Oregon no longer reported FRPL data to the CCD and not all states have CCD FRPL rates that track trends in state poverty (Spiegel, 2023).

How, then, should policymakers and researchers who lack access to detailed household income data measure the concentration of economic disadvantage in public schools? Our findings from Oregon indicate that school-level measures of the proportion of students enrolled in SNAP provides a highly valid measure of school economic disadvantage. We find that

temporal trends in SNAP enrollment rates parallel trends in our benchmark measure of student economic disadvantage. Further, we find that these two measures consistently correlate quite highly across the period for which we have data. Finally, we find that the correspondence between SNAP enrollment rates and our Benchmark (SNAP/IRS 3 year) measure holds equally well for Oregon public schools across the distribution of school economic disadvantage distribution. Notably, Oregon has a relatively high SNAP take up rate, and it is possible that SNAP-based measures might not fare as well in states where the barriers to SNAP participation for eligible families are higher.

These findings suggest that direct certification measures based on enrollment in SNAP and other means-tested federal programs may be accurate broadly, though further analysis of their reliability given variation in take-up across states is warranted. While direct certification measures are not currently publicly available for all U.S. public schools, a growing number of states make them available via the CCD and it is possible for all states to do so. Notably, SNAP enrollment reflects a different family income threshold than FRPL enrollment, so switching from one measure to the other requires attention to the corresponding level change in rates of economic disadvantage.

We find similarly encouraging evidence regarding the use of the Urban Institute MEPS measure. We note, however, that the Urban Institute MEPS measure is designed as an adjustment for publicly-available data, rather than an alternative source of data. In the Oregon context, the Urban Institute MEPS is calculated based on the relatively high-performing FRPL data reported in the CCD. It is not clear whether the Urban Institute MEPS measure would perform as well if it were calculated based on data in other states that was less highly correlated with economic disadvantage. Conversely, it is possible that the Urban Institute MEPS approach could help

further improve the validity of school economic disadvantage measures in contexts where direct certification data might not be as strong of a measure. Thus, the extent to which these findings hold for other states is an area for future research.

Our analyses proceed from the assumption that compensatory school policies intend to target resources at schools based on their enrolled concentration of students from economically disadvantaged families. While this focus is an explicit component of many educational policies, including Title I of the ESEA and many state and local school finance plans, we recognize that it is a policy choice. In the future it may be desirable to broaden our view, redirecting attention from the dichotomous and unidimensional notion of student economic disadvantage and toward richer, more multidimensional conceptions of student advantage and disadvantage (Singer, 2023). More sophisticated measures of student household economic status might consider household income as a continuous variable or incorporate local or regional cost of living data to better approximate household spending power. Further, in light of growing evidence regarding the role of wealth in the intergenerational reproduction of educational advantage (Hallsten and Pfeffer, 2017; Pfeffer and Killewald, 2018), policymakers and researchers may build wealth into their consideration of household economic status, though such an undertaking would likely necessitate new data-collection efforts. Alternatively, researchers and policymakers may attempt to broaden their view to consider non-economic factors such as household composition, educational attainment, occupational status, and availability of learning opportunities. Our analyses are not designed to speak to these broader constructs, but future research should.

What our research makes clear, however, is that commonly used measures of school economic disadvantage differ appreciably in their capacity to capture the variation in the proportion of students who reside in economically disadvantaged households across Oregon

public schools. These findings validate growing concerns over the quality of widely available measures of school economic disadvantage. At the same time, however, our findings indicate that direct certification data can perform well.

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[1] Throughout this paper, we use the terms "economic disadvantage" and "low-income" interchangeably. At several points in the manuscript, we make specific reference to the federal poverty line, a construct that is defined around a family's ability to secure a minimal standard of living. Since this construct is dated and does not account for regional variation in costs of living, scholars debate its validity (Hauser, 1994). Our discussion does not take a stand on this issue, but our use of the terms "economic disadvantage" and "low-income" rely on thresholds defined around the federal poverty line.

[2] Before summarizing these general strategies that are used or are proposed for measuring school economic disadvantage, we note that under Title I of the ESEA, the federal government allocates supplementary funds to districts based on data generated by the U.S. Census Bureau's Small Area Income and Poverty Estimates Program (SAIPE). SAIPE uses a combination of: county-level American Community Survey (ACS) data, summary data from federal income tax returns; SNAP benefits data, and data from the decennial Census. With these data, SAIPE estimates the percentage of children aged 5-17 residing within the district's geographical boundaries whose household income falls below the poverty line. Since these measures are largely derived from data reported by families sampled to participate in the ACS, they are not available at the school level. As such, the SAIPE data are beyond this paper's analytic scope.

[3] Families must demonstrate household income less than 130 percent of poverty and have less than $2250 in cash savings to enroll in SNAP. TANF eligibility rules vary across states, but are typically more restrictive than SNAP eligibility rules.

[4] Through an additional program called BlindSIDE, NCES has partnered with states to provide policymakers with a spatially interpolated school economic disadvantage measure based on student addresses, rather than school addresses (Institute of Education Sciences, 2021). While we

are unable to assess the properties of this proposed measure here, we note that it has a greater degree of face validity than the existing SNP measure. In FY2019, multiple State Departments of Education received additional funds from the State Longitudinal Data Systems Grant Program to help the National Center for Education Statistics to test this new Spatially Interpolated Demographic Estimates (SIDE) poverty measure. States participating in the testing of the SIDE measure collected and cleaned student addresses and obtained poverty estimates using the SIDE program and compared this measure with existing measures of poverty and other academic indicators, including FRPL data and test scores.

[5] For a detailed description of the Person Identification Validation System, see Wagner and Layne (2014).

[6] SNAP eligibility falls at 130% federal poverty, but students occasionally only briefly enrolled in SNAP during the school year may have income in excess of 185% federal poverty. However, our analysis suggests that these students overwhelmingly fall below the 185% threshold, so this measure still targets that threshold though it may not perfectly comply with it.

[7] For more information about the *Census-FNS-ERS Joint Project,* see the webpage: https://www.ers.usda.gov/topics/food-nutrition-assistance/food-assistance-data-collaborative-research-programs/snap-and-wic-administrative-data/

[8] Using SNAP enrollment directly instead of state-reported direct certification data avoids potential issues that arise due to matching issues in state administrative data systems. In school years 2017-18 and 2018-2019, the USDA estimates that 98 percent of SNAP-participant children were directly certified for free school meals. This rate was lower in Oregon, where 88 percent of children enrolled in SNAP were directly certified for free school lunch (USDA, 2021). Broadly speaking, eligible students might not be directly certified due to issues in the data linkage

procedures between SNAP and NLSP staff, or due to the timing of direct certification efforts compared with SNAP enrollment. Though, a 2011 USDA rule alleviates this concern by requiring schools to conduct direct certification matching efforts at least three times per year (USDA, 2011).

[9] There are different potential options for translating ACS income and poverty measures into school economic disadvantage rates. We found this approach superior to designating students in a binary fashion as either low or high income based on their neighborhood economic disadvantage rate or neighborhood average income.

[10] To do so, we first multiplied the provided standard errors of the NCES-SNP measure by five and calculated the proportion of the area under the normal curve that lies below 185 percent federal poverty, given the NCES-SNP mean and standard error.

[11] Since SNAP enrollment data contribute to our Benchmark (SNAP/IRS 3 year) measure, some of this correlation is mechanical. However, an alternative version of our benchmark measure based exclusively on the proportion of students whose families report income below 185 percent of the poverty line in their tax filings – referred to in Table 1 as the Benchmark (IRS 1 year) - also correlates very highly with the proportion enrolled in SNAP measure, a correlation of 0.942.

[12] The modified MEPS measure has a lower correlation with the benchmark than the original MEPS, with a correlation of .865.

[13] One reason that neighborhood-based measures might fare poorly is if large numbers of students at the school do not live in the neighborhood surrounding the school, as might happen if many students attend charter schools. Oregon is relatively average in terms of its charter school enrollment (5.7% of students attended charter schools in Oregon and 6.0% nationally in 2016-2017, see Wang, Rathbun, and Musu, 2019). Although charter schools are not the only kind of

choice available in Oregon, this suggests that the relatively low correlation of the NCES-SNP measure here is not due to an unusually large amount of school choice in Oregon.

[14] We plot binned scatterplots due to privacy constraints with data housed at the U.S. Census Bureau.

[15] It is also possible that SNAP overcounts economic disadvantage in some schools due to provisions in SNAP enrollment that allow families to remain enrolled without recertifying every year or that over-enrollment might be an artifact of the timing of income measurement.

[16] One might be worried that the high correspondence between the SNAP measure and the Benchmark (SNAP/IRS 3 year) is attributable to a mechanical correlation driven by the use of SNAP to construct the Benchmark (SNAP/IRS 3 year) measure. To assess this possibility, we looked at the correspondence of the SNAP based measure across the poverty distribution using an alternative benchmark – Benchmark (IRS 1 year) - that relies only on IRS data and find a similarly high degree of correspondence. Thus, we conclude that the high correspondence is not due to a mechanical correlation.

Table 1. List of measures of school economic disadvantage used in analyses

Name	*Source*	*Measure*	*Notes*
Benchmark measures			
Benchmark (SNAP/IRS 3 year)	Census (IRS/SNAP) 2007-08 through 2016-17	Proportion enrolled in SNAP at any point during relevant school year or with documented family income below <185% poverty	Students are designated low income if they (a) are enrolled in SNAP at any point during relevant school year or (b) have family income less than <185% poverty in their family's most recent IRS 1040 filling (within prior three tax years).
Benchmark (IRS 1 year)	Census (IRS) 2009-10 through 2016-17	Proportion with documented family income below <185% poverty	Students are designated low income if they have family income less than <185% poverty in their family's IRS 1040 from the most recent calendar year.
FRPL enrollment measures			
CCD FRPL	NCES - CCD 2009-10 through 2016-17	Proportion enrolled in FRPL	Based on state's reports to the NCES.
ODE Economic Disadvantage	Oregon Department of Education 2009-10 through 2016-17	Proportion flagged as "economically disadvantaged" based on NSLP enrollment and other factors	Other factors include children experiencing homelessness as defined by McKinney-Vento Homeless Assistance Act. In the 2015-2016 school year, all students are flagged as "economically disadvantaged" if they attend schools that provide free lunch to all regardless of economic status. This includes schools that are participating in CEP other universal meals programs.
Direct certification measure			
SNAP	Census (SNAP) 2009-10 through 2016-17	Proportion enrolled in SNAP (proxy for direct certification)	Received any SNAP benefits during school year (e.g., for 2017, enrollment between August 2016 and July 2017), excluding students who only receive SNAP via summer school-based program.
Alternative candidate measures			
ACS Block Group	ACS 2009-10 through 2016-17	Proportion <185% poverty in student's block group	Defined as the proportion of children ages 5-17 in households with income below <185% poverty in the student's block group.
Urban Institute MEPS	Urban Institute 2013-14 through 2018-19	Predicted poverty rate from SAIPE and CCD	This measure uses a linear mixed effects model to predict school-level estimates of economic disadvantage based on average household poverty rates for families with children in the districts in which schools are located.
NCES-SNP	NCES 2016-17	Income-to-poverty ratio	Based on the average income of households in the immediate vicinity of the school using ACS data.
Transformed NCES-SNP	NCES and author calculations 2016-17	Proportion <130% poverty	Transformation of the SNP measure designed to capture the share of students at or below 130% poverty for each school (see Fazlul et al., 2021).

Source: 2010-2017 ODE, IRS 1040, OR SNAP
DRB Approval Numbers: CBDRB-FY2022-CES010-023, CBDRB-FY21-CES014-017.

Table 2. Enrollment weighted descriptive statistics of analytic sample

	Analytic sample	
	Mean	*Std. Dev.*
Benchmark measures		
Benchmark (SNAP/IRS 3 year)	0.506	0.185
Benchmark (IRS 1 year)	0.433	0.171
Traditional FRPL Enrollment Measures		
CCD FRPL	0.522	0.211
ODE Economic Disadvantage	0.539	0.242
Direct certification measure		
SNAP*	0.345	0.164
Alternative candidate measures		
ACS Block Group*	0.327	0.093
Urban Institute MEPS+	0.176	0.075
NCES-SNP (Income-to-poverty ratio)+	304.9	140.5
Transformed NCES-SNP+	0.399	0.106
Demographic characteristics		
Proportion Black	0.024	0.048
Proportion White	0.641	0.191
Proportion Hispanic	0.221	0.174
Proportion Multi-ethnic	0.048	0.031
Proportion American-Indian	0.016	0.041
Proportion Asian	0.049	0.063
Proportion ESL	0.227	0.219
School Enrollment	733.5	520.2
N (School-years)	8500	

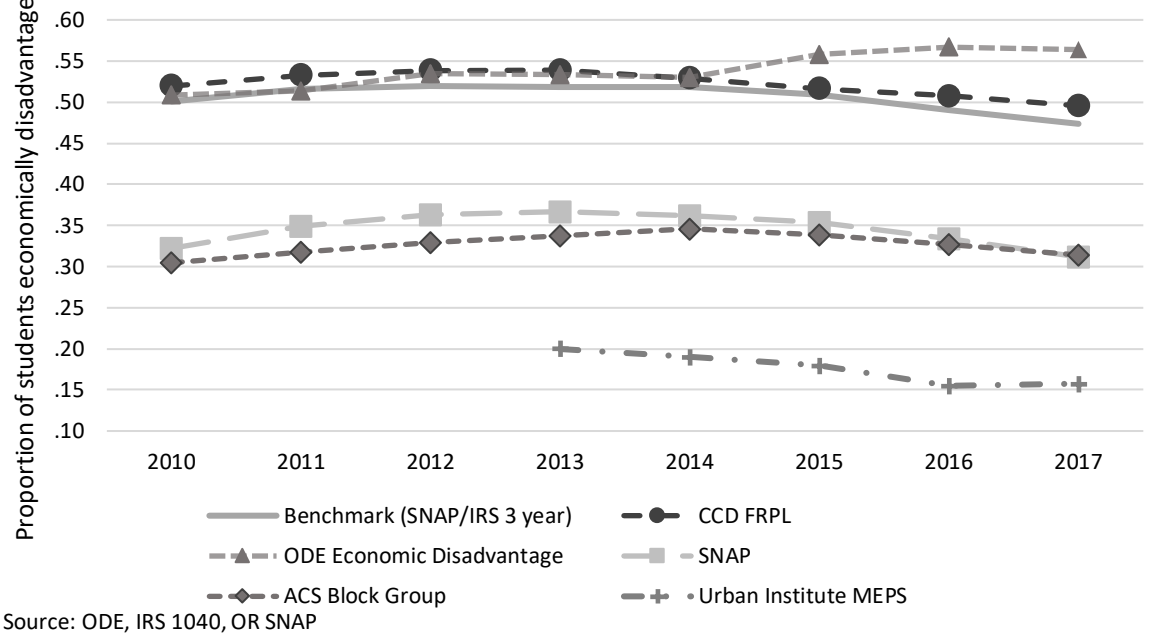
Source: 2010-2017 ODE, IRS 1040, OR SNAP
DRB Approval Numbers: CBDRB-FY2022-CES010-023, CBDRB-FY21-CES014-017.

Notes. Statistics are enrollment weighted.

*Available for >99% of school-year observations in the full sample, but missing for a small number of school-years. Measures with suppressed values in the full sample have many more missing values. By construction, all measures are available for all schools in the longitudinal sample.

+ Urban Institute MEPS measures are available 2013-2017; the NCES-SNP measure in the analysis is from 2017.

Figure 1. Trends in student economic disadvantage in Oregon across measures, 2010-2017



Source: ODE, IRS 1040, OR SNAP

DRB Approval Numbers: CBDRB-FY2022-CES010-023, CBDRB-FY21-CES014-017.

Table 3. Correlations between benchmark measures and candidate measures, 2016-2017 school year

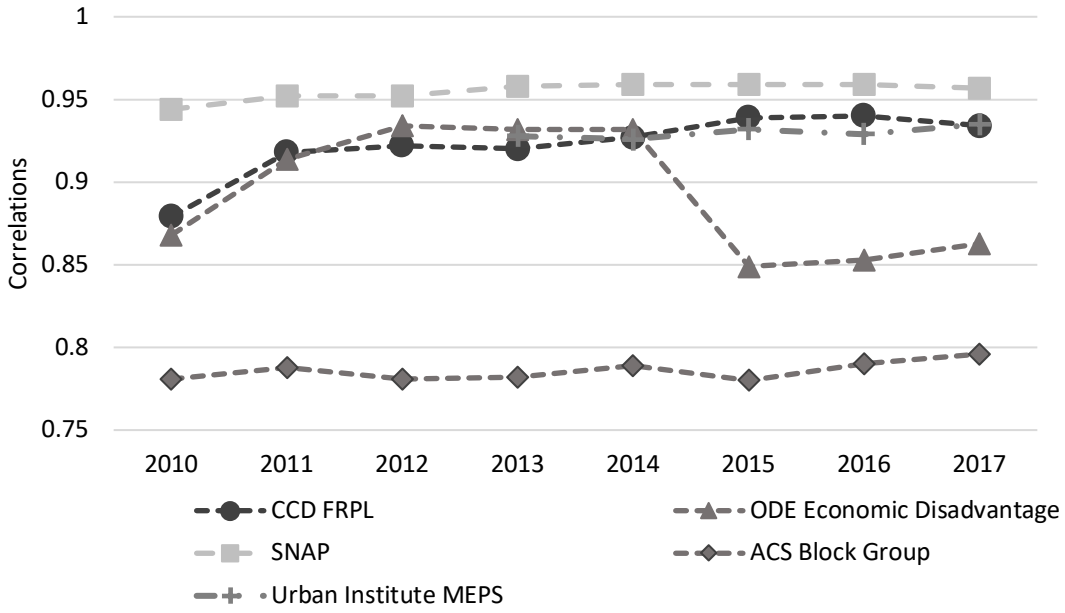
	Benchmark (SNAP/IRS 3 year)	*Benchmark (IRS 1 year)*
Benchmark measures		
Benchmark (SNAP/IRS 3 year)	1	
Benchmark (IRS 1 year)	0.991	1
Traditional FRPL Enrollment Measures		
CCD FRPL	0.934	0.923
ODE Economic Disadvantage	0.863	0.859
Direct certification measure		
SNAP	0.957	0.942
Alternative candidate measures		
ACS Block Group	0.796	0.796
Urban Institute MEPS	0.935	0.929
NCES-SNP (Income-to-poverty ratio)	-0.694	-0.679
Transformed NCES-SNP	0.661	0.653
N Schools	1,100	

Source: ODE, IRS 1040, OR SNAP
DRB Approval Numbers: CBDRB-FY2022-CES010-023, CBDRB-FY21-CES014-017.

Note. Statistics are not enrollment weighted.
Statistics include schools from the final year of the analytic sample (2016-2017) for which there is coverage from all measures. We lose approximately 300 schools with small enrollments driven in large part to whether schools are represented in the NCES data sources like the CCD and SNP measures. The analytic sample offers a reasonable snapshot of the universe of schools in the ODE data (see Appendix Table 1).

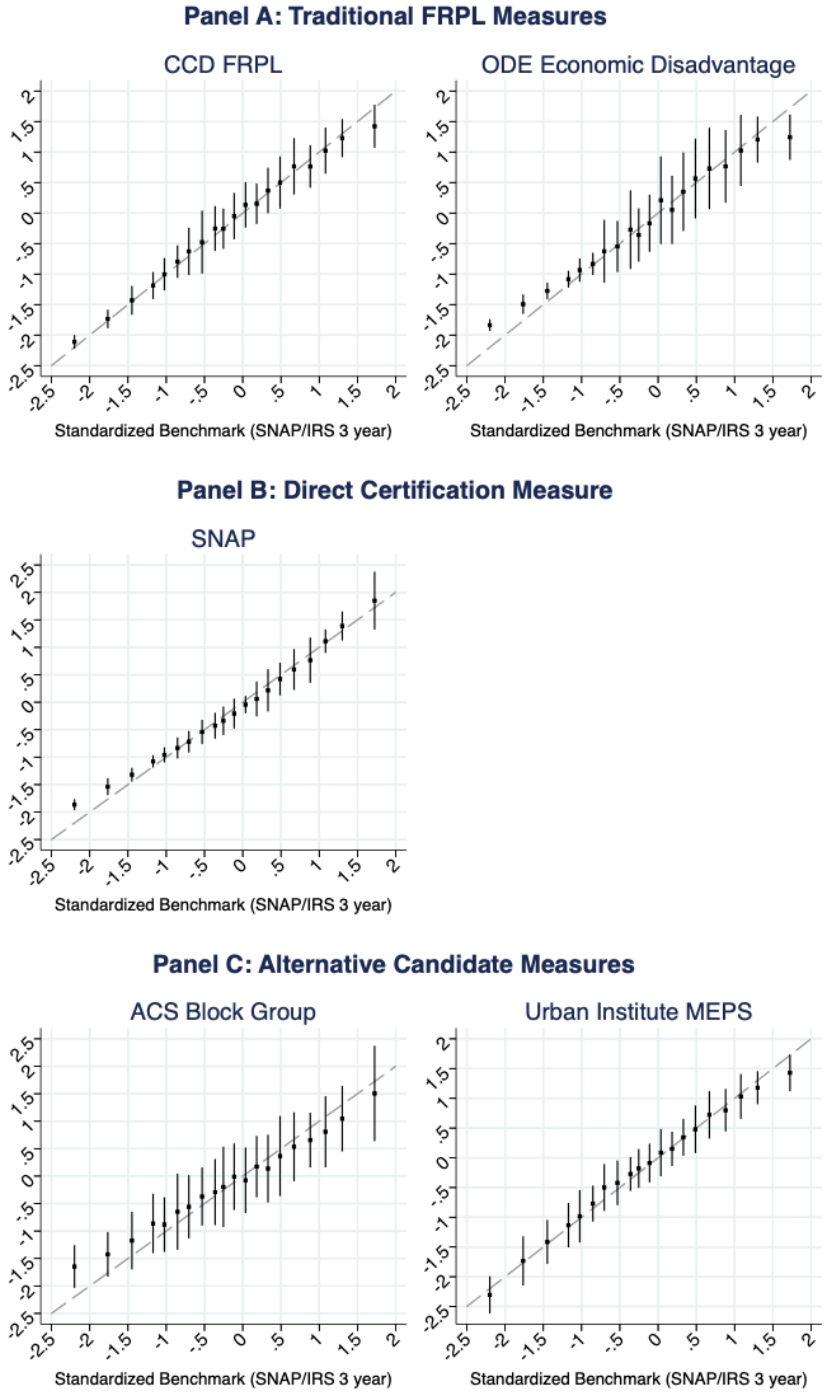
The Urban Institute also released a MEPS modified measure which they recommend to be used for analyses in large school districts with large school enrollments. Because we are conducting a state-wide analysis with schools of varying enrollment sizes, we do not focus on the MEPS modified measure in the paper. Consistent with Urban Institute's recommendation, the MEPS modified measure has a lower correlation with the Benchmark (SNAP/IRS 3 year) than the MEPS, with a correlation of .865.

Figure 2. Trends in the correlation between the Benchmark (SNAP/IRS 3 year) and candidate measures, 2010-2017



Source: ODE, IRS 1040, OR SNAP
DRB Approval Numbers: CBDRB-FY2022-CES010-023, CBDRB-FY21-CES014-017.

Figure 3. Relationship between candidate measures and Benchmark (SNAP/IRS 3 year) across the distributions



Source: ODE, IRS 1040, OR SNAP
DRB Approval Numbers: CBDRB-FY2022-CES010-023, CBDRB-FY21-CES014-017.
Note. Axes ranges differ slightly to optimize visibility. Data is from academic year 2016-2017. Measures are standardized, with higher values indicating higher economic disadvantage and lower values indicating lower economic disadvantage.

Appendix Table 1. Enrollment weighted descriptive statistics of analytic sample and the universe of schools in ODE data, 2009-10 through 2016-17

	Universe of schools in ODE data		Analytic sample	
	Mean	*Std. Dev.*	*Mean*	*Std. Dev.*
Benchmark measures				
Benchmark (SNAP/IRS 3 year)	0.508	0.186	0.506	0.185
Benchmark (IRS 1 year)	0.435	0.172	0.433	0.171
Traditional FRPL enrollment measures				
CCD NSLP	-	-	0.522	0.211
ODE Economic Disadvantage	0.530	0.245	0.539	0.242
Direct certification measure				
SNAP*	0.346	0.165	0.345	0.164
Alternative candidate measures				
ACS Block Group*	0.327	0.092	0.327	0.093
Urban Institute MEPS+	-	-	0.176	0.075
NCES-SNP (Income-to-poverty ratio)+	-	-	304.9	140.5
Transformed NCES-SNP+	-	-	0.399	0.106
Demographic characteristics				
Proportion Black	0.025	0.052	0.024	0.048
Proportion White	0.646	0.190	0.641	0.191
Proportion Hispanic	0.216	0.173	0.221	0.174
Proportion Multi-ethnic	0.049	0.032	0.048	0.031
Proportion American-Indian	0.016	0.041	0.016	0.041
Proportion Asian	0.048	0.062	0.049	0.063
Proportion ESL	0.220	0.219	0.227	0.219
School Enrollment	720.5	564	733.5	520.2
N (School-years)	12000		8500	

Source: ODE, IRS 1040, OR SNAP
DRB Approval Numbers: CBDRB-FY2022-CES010-023, CBDRB-FY21-CES014-017.

Notes. Statistics are enrollment weighted.
*Available for >99% of school-year observations in the full sample but missing for a small number of school-years. Measures with suppressed values in the full sample have many more missing values. By construction, all measures are available for all schools in the analytic sample.

+ Urban Institute MEPS measures are available 2013-2017; the NCES-SNP measure used in the analysis is from 2017.

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